

- c) Smartphone B has more RAM, more ROM, and a higher resolution camera than Smartphone A $\rightarrow$ False.
- d) If Smartphone B has more RAM and more ROM than Smartphone C, then it also has a higher resolution camera $\rightarrow$ False.
- e) Smartphone A has more RAM than Smartphone B if and only if Smartphone B has more RAM than Smartphone A $\rightarrow$ False

9. Suppose that during the most recent fiscal year, the annual revenue of Acme Computer was 138 billion dollars and its net profit was 8 billion dollars, the annual revenue of Nadir Software was 87 billion dollars and its net profit was 5 billion dollars, and the annual revenue of Quixote Media was 111 billion dollars and its net profit was 13 billion dollars. Determine the truth value of each of these propositions for the most recent fiscal year.

- a) Quixote Media had the largest annual revenue $\rightarrow$ False
- b) Nadir Software had the lowest net profit and Acme Computer had the largest annual revenue $\rightarrow$ True
- c) Acme Computer had the largest net profit or Quixote Media had the largest net profit $\rightarrow$ True
- d) If Quixote Media had the smallest net profit, then Acme Computer had the largest annual revenue $\rightarrow$ True
- e) Nadir Software had the smallest net profit if and only if Acme Computer had the largest annual revenue $\rightarrow$ True

10. Let p and q be the propositions
 p : I bought a lottery ticket this week. q : I won the million dollar jackpot

- Express each of these propositions as an English sentence
- a) $\neg p$ $\rightarrow$ I did not buy a lottery ticket this week.
- b) $p \vee q$ $\rightarrow$ I bought a lottery ticket this week or I won the million dollar jackpot.
- c) $p \rightarrow q$ $\rightarrow$ If I bought a lottery ticket this week, then I won the million dollar jackpot.
- d) $p \wedge q$ $\rightarrow$ I bought a lottery ticket this week and I won the million dollar jackpot.